Platform and set-up with FreeRunner

- *Platform (PC):* Dell Studio 1555 laptop running Ubuntu 10.04.1 LTS, with Sun Microsystems Java 6 Update 18 (specifically installed via Synaptic, package named sun-java6-jre).
- *Platform (phone):* Openmoko FreeRunner GTA02, running Android 1.5 Cupcake and I also tried the emulator packaged with the App Inventor Extras software. (I installed the version distributed by the AndroidOnFreeRunner project, which took over from the Koolu attempt to port Android to the FreeRunner. The project shifted from the Openmoko site to Google Code. Installation instructions are at http://goo.gl/QxxM. Find more information about the project on the same wiki.)
- App Inventor Extras Software: *appinventor-extras_1.0-2_all.deb*
- Browser: Firefox 3.6.10 with the NoScript extension disabled as advised.

The instructions to set up your phone and computer are at http://goo.gl/ebax, so I will not repeat them here.

For the *Install the App Inventor Extras Software* section, since I am running Ubuntu, I visited the 'Instructions for GNU/Linux' (http://goo.gl/61Ab) and downloaded *appinventor-extras_1.0-2_all.deb* from the link. I installed it with GDebi (double-clicked the downloaded deb file in Nautilus).

As instructed in the *Set up your phone* section, I found and set *Unknown Sources*, *USB Debugging*, *Stay Awake* and *Orientation*. However, where they say: "*Make sure your phone is unlocked — that is, ready to make a call or run an app. Connect the cable from the phone to the computer's USB port. You should see two new notifications appear on the phone, with icons in the status bar: A 'USB connected' notification that the phone is connected to the computer via USB /and/ A 'USB debugging connected' notification that the phone has USB debugging turned on*"—here, I found that I was not able to see these notifications.

After re-checking the whole procedure, and still failing to see the notifications, I was stymied, and tried the *adb devices* command (the Android debug bridge, installed by the App Inventor Extras Software) to see if it found the FreeRunner. But no devices were detected.

A Google search led me to http://goo.gl/Nf79, where I found out what was wrong. The USB networking connection (the *eth1* interface on my system) was auto-configured in Network Manager to automatically connect—but to request an IP address via DHCP! Android doesn't supply that, of course. I changed that to Manual, and assigned the IP address (PC side) as 192.168.0.200/24, since Android on FreeRunner sets the phone's address to 192.168.0.202. The *Connect automatically* check-box wouldn't stay checked, so I used a left-click on the Network Manager icon in the notification area to get to the list of available connections, where I enabled the Wired connection for the FreeRunner.

After it came up, I ran an *adb kill-server* command, followed by:

```
$ ADBHOST=192.168.0.202 adb devices
List of devices attached
emulator-5554 device
```

Strange name, but I was seeing the FreeRunner! To make sure, I tried to view the phone's log:

```
$ export ADBHOST=192.168.0.202
$ adb shell "logcat"
[snip several hundred lines]
I/ActivityManager( 870): Displayed activity com.android.settings/.DeviceInfoSettings: 879 ms
D/lights ( 870): Battery charging or low buttons /sys/class/leds/gta02-power:orange ON (255)
D/dalvikvm( 870): GC freed 13828 objects / 680744 bytes in 191ms
I/ActivityManager( 870): Starting activity: Intent { action=android.intent.action.MAIN comp={com.android.settings/com.android.
settings.ApplicationSettings} }
I/ActivityManager( 870): Displayed activity com.android.settings/.ApplicationSettings: 609 ms
I/ActivityManager( 870): Starting activity: Intent { action=android.intent.action.MAIN comp={com.android.settings/com.android.
settings.DevelopmentSettings} }
I/ActivityManager( 870): Displayed activity com.android.settings/.DevelopmentSettings: 805 ms
D/lights ( 870): Battery charging or low buttons /sys/class/leds/gta02-power:orange ON (255)
```

I had let the phone's battery drain down to zero, and it was recharging—and besides, I didn't have any emulator running—so now I knew that *adb* was indeed showing me a connection to the phone.

Note: *For phones running later versions of Android, you will probably see those notifications I didn't get, but for Android 1.5 Cupcake, especially the port on the FreeRunner, ignore the App Inventor set-up page's points about notifications and icons in the status bar. If adb is connecting to the phone and can give you a log output, it should work.*

The next big step was connecting App Inventor to the phone. This is documented at http://goo.gl/ZXa2, and again, I won't repeat any of it. I proceeded with the steps—creating a new project, opening the Blocks Editor, and clicking the *Connect to phone* button. However, it had been quite a while since my last *adb* communication with the phone, and so this failed, with an error message that the Blocks Editor could not detect any connected devices. Returning to my terminal window, I ran:

```
$ adb get-state
adb server is out of date. killing...
* daemon started successfully *
device
```

Another *adb shell 'logcat'* returned the previous log (plus much more). With communication with the phone re-established, I went back to the Blocks Editor, and clicked *Connect to phone* again. This time, after about 8 to 12 seconds of a 'Communicating' message, with an animated icon, it established the connection and opened the (to be designed) application preview on the phone's screen.

With this working, I proceeded to the tutorials, as advised at the end of the page. However, my experiences with reconnecting the phone with Blocks Editor tended to vary a little, so both for convenience, and for the couple of extra points, I've included another box with a check-list for reconnecting the phone.